LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 9 Test (GSEB pattern): Light – Reflection and Refraction

Chapter / Topic: Light – Reflection and Refraction

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. Which of the following can make a parallel beam of light when light from a point source is incident on it? **[1]**

(a) Concave mirror and convex lens
(b) Convex mirror and concave lens
(c) Two plane mirrors at 90° to each other
(d) Concave mirror and concave lens

Q2. The radius of curvature of a convex mirror is 32 cm. Its focal length is: **[1]**

(a) −64 cm (b) −16 cm
(c) +16 cm (d) +32 cm

Q3. The image formed by a concave mirror is virtual, erect and larger than the object. Where is the object placed? **[1]**

(a) Between F and C (b) At C
(c) Beyond C (d) Between the pole P and F

Q4. Which of the following is used as a magnifying glass? **[1]**

(a) Concave lens (b) Plane glass sheet
(c) Prism (d) Convex lens

Q5. The focal length of a lens is +50 cm. Its power is: **[1]**

(a) −2 D (b) +0.5 D
(c) +2 D (d) +50 D

Q6. Light travels fastest in: **[1]**

(a) vacuum (b) glass
(c) water (d) diamond

Q7. The refractive index of glass is 1.5. The speed of light in glass is (speed of light in vacuum = 3×10^8 m/s): **[1]**

(a) 1.5×10^8 m/s (b) 2×10^8 m/s
(c) 3×10^8 m/s (d) 4.5×10^8 m/s

Q8. The magnification produced by a plane mirror is: **[1]**

(a) −1 (b) +1
(c) zero (d) infinity

Q9. The mirror used as a rear-view mirror in vehicles is: **[1]**

(a) convex (b) plane

(c) concave (d) any of these

SECTION B

6 marks

Q10. Define the principal focus of a concave mirror. The radius of curvature of a spherical mirror is 20 cm. What is its focal length? **[2]**

Q11. Define the power of a lens and give its SI unit. Find the power of a convex lens of focal length 40 cm. **[2]**

Q12. A ray of light travelling in air enters water at an angle. Does it bend towards or away from the normal? Why? **[2]**

SECTION C

6 marks

Q13. An object 4 cm high is placed 15 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) Find the position of the image. **[2]**

(ii) Find the size and nature of the image. **[1]**

Q14. An object 5 cm high is placed 20 cm in front of a concave mirror of focal length 10 cm. **[3]**

(i) Find the position of the image. **[2]**

(ii) Find the size and nature of the image. **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. The refractive index of a medium tells us how much light slows down in it compared with a vacuum. Some values are: water 1.33, kerosene 1.44, crown glass 1.52 and diamond 2.42. **[4]**

(i) In which of these four media does light travel fastest? **[1]**

(ii) Which of these is optically the densest? **[1]**

(iii) Calculate the speed of light in diamond. (Speed of light in vacuum = 3×10^8 m/s.) **[2]**

~ All the best! ~